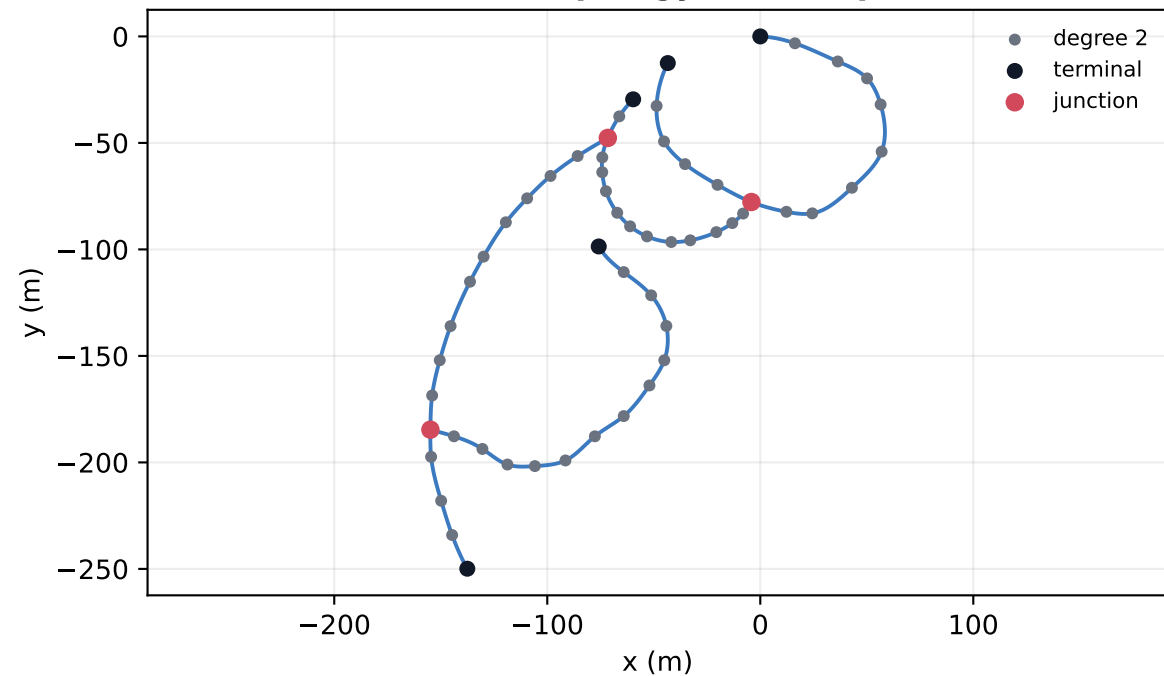
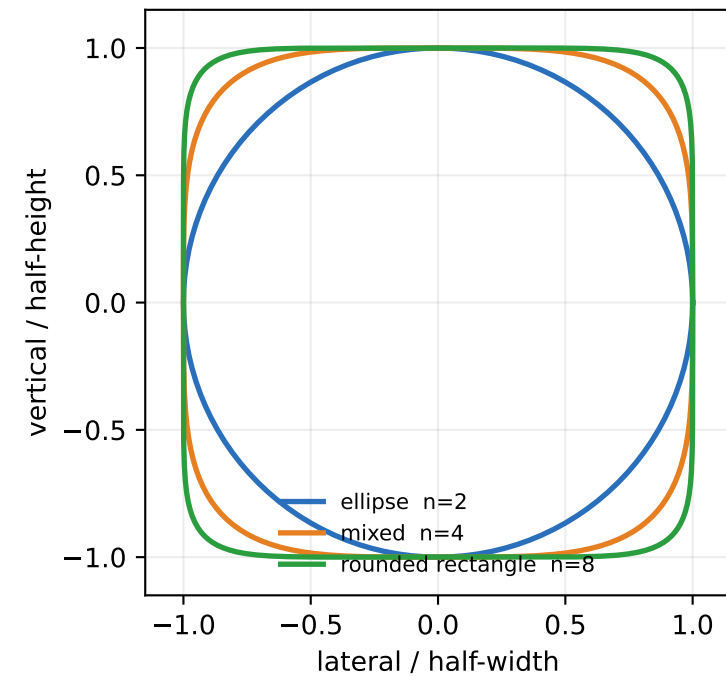


Procedural tunnel construction becomes causal primitive-relation supervision

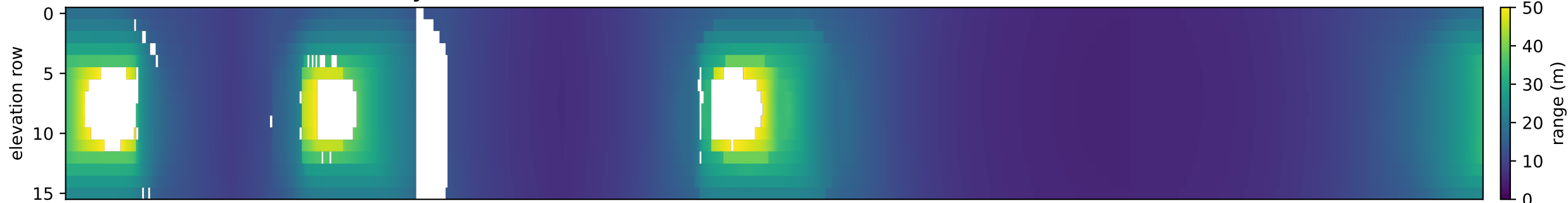
(a) Generated tunnel topology and swept axes



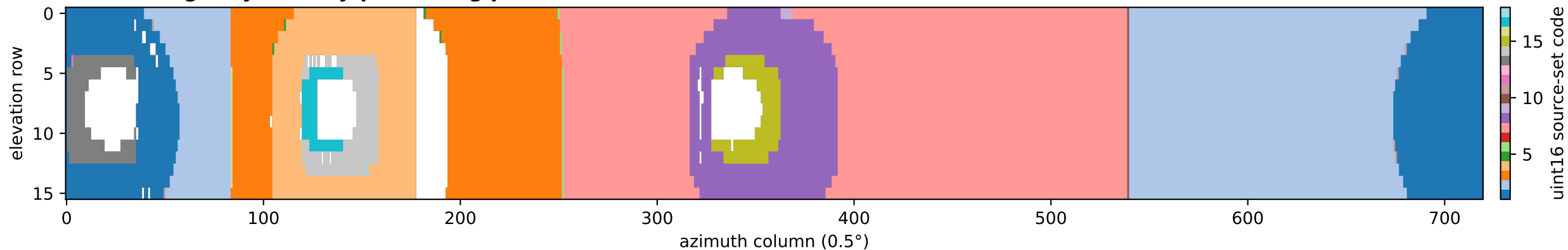
(b) One continuous cross-section family



(c) Causal LiDAR frame used by the student



(d) Training-only identity-preserving primitive source-set code



80 topology parents · 240 paired geometry tasks · 757,290 scans · 564,378 five-frame windows · 24,117 realized primitives
8,779,620 directed attachments · 3,023,794 disconnected overlaps · 538,391 temporal dustbins · construction identity is never a deployed-model input